

Lunar phases and zodiac signs do not influence quality of radical cystectomy—a statistical analysis of 452 patients with invasive bladder cancer

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Abstract

Objective To determine the influence of the lunar phases and the position of the moon in the zodiac on the frequency of complications and the survival of bladder cancer patients after radical cystectomy. It has been postulated that radical cystectomy performed during the waxing moon, or particularly at full moon, or at the zodiac sign Libra is associated with a poorer outcome.

Material and methods We tested this hypothesis by evaluating the progression-free survival, the complication rate and the re-operation rate for 452 consecutive patients after radical cystectomy. In this retrospective review, the dates of surgery

were allocated to the lunar phases and the zodiac signs. Based on these classifications, the patients were placed in groups which combined the lunar phase laws and differentiated between evidently unfavorable (full moon or waxing moon and/or the zodiac sign Libra; assigned to group 1) and favorable periods for surgery (new moon or waning moon and other signs of the zodiac apart from Libra; assigned to group 2). The mean follow-up was 49 months (range 0–158 months).

Results A total of 244 patients (54%) were operated during an unfavorable period (group 1) and 208 (46%) patients during the auspicious period (group 2). The mean age, gender and kind of urinary derivation did not differ significantly in the two groups. Pathological tumor stages were evenly distributed according to the lunar phase groups ($P = 0.713$). We found no significant differences in the perioperative mortality rates, early re-operation rates, early complications, and late complications across the two groups. No significant differences in progression-free survival were observed when timing of cystectomy during the lunar cycle was considered ($P = 0.231$).

Conclusions Our analysis demonstrated no predictable influence of the lunar phase on survival or complications. Although this was not a prospective randomized trial, the statistical magnitude of the results do not support any recommendations

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for scheduling patients for radical cystectomy at any particular day of the lunar phase.

Keywords Cystectomy · Lunar phases · Postoperative complications · Progression-free survival · Zodiac signs

Introduction

The lunar cycle has been credited with influencing various human phenomena. The rationale for these proposed effects have been attributed to differences in available moonlight, barometric pressure (weather conditions), geomagnetic and gravitational variations, and solar corpuscular radiation, among others [1]. The potential influence of lunar phases and zodiac signs on human behavior and the incidence of disease have been widely discussed by the lay press [2]. Several reports have been published focussing on the moon's influence on suicides [3, 4], cardiopulmonary resuscitations [1], birth rates [5, 6], renal colic [7], postoperative complications [8, 9], and survival after therapy [10]. Although most of these studies concerning the moon's influence on quality of treatment were negative or inconclusive, the reports in the press have led to serious concerns among patients and, to some extent, even among physicians. In fact, 10.5% of the German population believes in the effects of lunar phase on disease [11]. The number of patients who arrange planned operations according to the lunar cycle is on the increase. Especially since the attitude of the patient also decisively affects the progress of disease, superficially negating the fears of the patient with regard to the relevance of the lunar phases can have a negative effect on the surgical outcome. On the other hand, a brief delay to the timing for surgery may cause organizational problems, in addition to which numerous working groups have confirmed that especially where invasive cancer of the bladder is concerned, a time delay until surgery actually takes place is associated with a worsening prognosis [12, 13, 14].

According to the lunar phases, surgical interventions of any kind during a full moon, a waxing moon, and when the moon passes through the

sign of the zodiac for the physical region (urinary bladder = Libra) have a less favorable outcome [2, 8, 9]. The rationale for our test is the fact that no evidence-based statement is available from the international bibliography, as to whether the lunar phase and the sign of the zodiac has any effect on the rate of complications and tumor control at the time of a radical cystectomy.

In this study, the progress of 452 patients with bladder cancer was recorded, who had had a cystectomy during a 12-year period at three urological clinics. The lunar phases and the zodiac signs of all patients at the time of surgery were recorded, and subsequently correlated with the early and delayed complications, the re-operation rate and progression-free survival.

Material and methods

Patients

A total of 452 bladder cancer patients (all patients in stage cM0) with adequate information on the exact surgery dates and a complete documented history were evaluated. All patients underwent radical cystectomy during February 1992 and September 2004 at the urology clinics in Vivantes Hospital Berlin-Friedrichshain, Vivantes Hospital Berlin-Am Urban, and Carl-Thiem Hospital Cottbus. The histopathology of local tumor stage (pT stage) and lymph node invasion (pN stage) were described according to the 2002 International Union Against Cancer TNM classification (6th edition). Pathological stages were defined on the basis of the organ-confined and lymph node-negative tumor (pT0, pTis, pTa-pT2b; appropriated to OC,pN0), extravesical invasion and lymph node-negative tumor (pT3a-pT4b; appropriated to NOC,pN0), and also the lymph node-positive tumor (pT0-pT4, pN+; appropriated to pN+).

The study group included 347 male (76.8%) patients and 105 female (23.2%) patients producing a gender ratio of 3.3:1. The average patient age at cystectomy was 64.3 (35–84) years with a median of 64.9 years. Urinary diversion was carried out using an orthotopic ileum-neobladder in 192 patients (42.5%) and an ileum-conduit in 260 (57.5%).

The date of definitive surgery was allocated to the dates and times of the lunar phases and to the zodiac signs [15]. Categorization of the lunar phases was carried out in such a manner that the period of the full moon was considered to be ± 2 days around the precise day of the full moon (lunar phase 1). The new moon was defined in a similar manner (lunar phase 3). Additionally, patients were divided according to the period of the waning moon (defined as the time interval from lunar phase 1 to lunar phase 3; corresponded to lunar phase 2) and to the waxing moon (defined as the time interval from lunar phase 3 to lunar phase 1; appropriated to lunar phase 4). In line with the applicable lunar phase calendars, as well as in general according to the iatromathematical tradition, the urinary bladder was allocated to the sign of the zodiac Libra [2, 15]. A simple dichotomy of the patients took place between those who had a radical cystectomy carried out during the sign of the zodiac Libra, and those that were operated on at some other time (non-Libra). A further grouping of the patients then took place, which combined the lunar phase laws and differentiated between evidently unfavorable (full moon or waxing moon and/or the zodiac sign Libra; assigned to group 1) and favorable periods for surgery (new moon or waning moon and other signs of the zodiac apart from Libra; assigned to group 2).

Procedure, follow-up, and statistical analysis

During the first two postoperative years all patients were examined at 3-month intervals. This was extended to 6 months from the third postoperative year. Patients were subsequently examined every 12 months after the fifth postoperative year. This involved a full physical examination, abdominal sonography, and thorax X-rays as well as analysis of laboratory parameters. Intravenous urography was carried out after 3 months and thereafter at 12-monthly intervals. The use of CT, MRI, and bone scans was limited to individual cases.

The entire study group was observed for a mean follow-up of 49 (0–158) months (median

38 months). A total of 83.0% of the patients were observed for a further time of ≥ 12 months. The follow-up update was carried out in June 2005.

As potential dependent variables were recorded: (a) perioperative mortality (each death within 30 days after cystectomy), (b) early complications (each complication within 3 months after cystectomy), (c) late complications (>3 months after cystectomy), (d) early re-operation rate (each re-operation within 30 days after cystectomy), and (e) progression-free survival. Progression-free survival represented the time between radical cystectomy and the first recurrence of a tumor; patients, who died without any signs of tumor growth, were censored at the time of death. Survival times of patients who had survived without tumor progression were censored with the end of the study. The Kaplan–Meier method was used to determine survival probability and progression rates. The log-rank test was used to examine the differences between the groups on the basis of their progression-free survival rates. The prognostic value of the lunar phase-division was determined with aid of the Cox proportional hazard model. Significant differences in the distribution of the variables a–d (see above) in relation to the lunar phases and the zodiac sign groups were shown using the Chi-square and ANOVA tests. It was also finally tested as to whether fewer operations were carried out during the time frames that were deemed to be unfavorable according to the lunar phase laws (group 1), than during the other lunar phases or signs of the zodiac (group 2). This might indicate a certain amount of confidence, respectively an avoidance strategy among the patients. A *P*-value less than or equal to 5% was considered statistically significant. The SPSS® 13.0 statistical software system was used for calculations.

Results

Eighty three procedures were performed in full moon (18.4%), 146 (32.3%) in the waning moon, 78 (17.2%) in new moon, and 145 operations (32.1%) at waxing moon. There was no difference in the distribution of operations with regard to the lunar phases (lunar phases 1 + 4 and lunar phases 2 + 3 with 50.5% vs. 49.5%, respectively). During

Table 1 Patient characteristics of 452 bladder cancer patients according to the lunar phase. *Legend:* OC = organ confined tumor, NOC = non-organ confined tumor/extravesical invasion

Patient characteristic	Lunar phase 1 (Full moon, <i>n</i> = 83)	Lunar phase 2 (Waning moon, <i>n</i> = 146)	Lunar phase 3 (New moon, <i>n</i> = 78)	Lunar phase 4 (Waxing moon, <i>n</i> = 145)	<i>P</i> -value
Mean age in years (SD)	64.6 (±10.6)	64.2 (±8.2)	65.4 (±7.8)	63.5 (±8.5)	0.472
Male gender	64 (77.1%)	111 (76.0%)	59 (75.6%)	113 (77.9%)	0.975
Ileum-neobladder	36 (43.4%)	65 (44.5%)	32 (41.0%)	59 (40.7%)	0.911
OC, pN0	39 (47.0%)	61 (41.8%)	50 (64.1%)	63 (43.4%)	0.032
NOC, pN0	21 (25.3%)	40 (27.4%)	9 (11.5%)	42 (29.0%)	
pN+	23 (27.7%)	45 (30.8%)	19 (24.4%)	40 (27.6%)	
Perioperative mortality	1 (1.2%)	3 (2.1%)	3 (3.8%)	4 (2.8%)	0.721
Early complications	36 (43.4%)	51 (34.9%)	32 (41.0%)	61 (42.1%)	0.523
Late complications	24 (28.9%)	62 (42.5%)	24 (30.8%)	49 (33.8%)	0.132
Early re-operation	6 (7.2%)	10 (6.8%)	3 (3.8%)	5 (3.4%)	0.453

the zodiac sign of Libra which was considered to be critical, 37 cystectomies (8.2%) were performed. This was close to the statistically expected range of 1/12 (8.3%). The distribution of operations during the lunar phases and signs of the zodiac was thus almost at parity and a dominant avoiding strategy regarding “unfavorable operating dates” was far more improbable.

Patient characteristics and pathological stages are presented for lunar phase, zodiac sign and lunar phase groups in Tables 1–3. We found no significant differences in the perioperative mortality rate and the 30-days-reoperation rate between the 4 lunar phases (Table 1), between the 2 zodiac sign groups (Table 2), and between the 2 lunar phase groups (Table 3). We further found no significant differences for early and late complications (Tables 1–3).

After 5 years the probabilities of progression-free survival and overall survival in the entire study population was 56.1% and 53.6%,

respectively. Organ-confined lymph node-negative tumors (*n* = 213, 47.1%) with a progression-free 5-year survival rate of 78.4% were statistically more significant (*P* < 0.001) than non-organ-confined lymph node-negative tumors (*n* = 112, 24.8%), which only had a progression-free 5-year survival rate of 42.3%. The progression-free 5-year survival rate for lymph node-positive patients (*n* = 127, 28.1%) was 29.0%. It was therefore significantly poorer than in lymph node-negative patients at the OC (*P* < 0.001) or NOC (*P* = 0.009) tumor stage (Fig. 1).

No significant differences of progression-free survival were observed when timing of cystectomy was allocated to the lunar phases (*P* = 0.102, Fig. 2) or zodiac signs (*P* = 0.269, Fig. 3). In addition to these results, Kaplan–Meier statistic did not reveal any impact of lunar phase groups on patients progression-free survival (*P* = 0.231, Fig. 4). Finally, the lunar phase group variable was entered as covariate in a Cox proportional

Table 2 Patient characteristics of 452 bladder cancer patients according to the zodiac sign (Libra versus non-Libra). *Legend:* OC = organ confined tumor, NOC = non-organ confined tumor/extravesical invasion

Patient characteristic	Zodiac sign Libra (<i>n</i> = 37)	Zodiac sign non-Libra (<i>n</i> = 415)	<i>P</i> -value
Mean age in years (SD)	66.1 (±9.8)	64.1 (±8.7)	0.190
Male gender	29 (78.4%)	318 (76.6%)	0.809
Ileum-neobladder	11 (29.7%)	181 (43.6%)	0.102
OC, pN0	19 (51.4%)	194 (46.7%)	0.835
NOC, pN0	9 (24.3%)	103 (24.8%)	
pN+	9 (24.3%)	118 (28.4%)	
Perioperative mortality	0	11 (2.7%)	0.316
Early complications	12 (32.4%)	168 (40.5%)	0.338
Late complications	10 (27.0%)	149 (35.9%)	0.279
Early re-operation	1 (2.7%)	23 (5.5%)	0.710

Table 3 Patient characteristics of 452 bladder cancer patients according to the lunar phase groups (group 1/unfavorable period versus group 2/auspicious period). *Legend:* OC = organ confined tumor, NOC = non-organ confined tumor/extravesical invasion

Patient characteristic	Group 1 (unfavourable period, <i>n</i> = 244)	Group 2 (auspicious period, <i>n</i> = 208)	<i>P</i> -value
Mean age in years (SD)	63.9 (± 9.4)	64.6 (± 7.9)	0.406
Male gender	189 (77.5%)	158 (76.0%)	0.707
Ileum-neobladder	100 (41.0%)	92 (44.2%)	0.486
OC, pN0	114 (46.7%)	99 (47.6%)	0.713
NOC, pN0	64 (26.2%)	48 (23.1%)	
pN+	66 (27.0%)	61 (29.3%)	
Perioperative mortality	5 (2.0%)	6 (2.9%)	0.566
Early complications	101 (41.4%)	79 (38.0%)	0.460
Late complications	79 (32.4%)	80 (38.5%)	0.177
Early re-operation	12 (4.9%)	12 (5.8%)	0.834

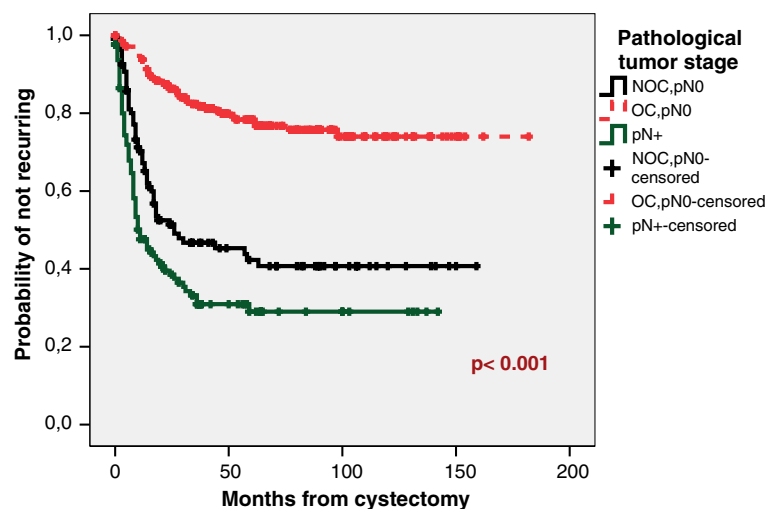
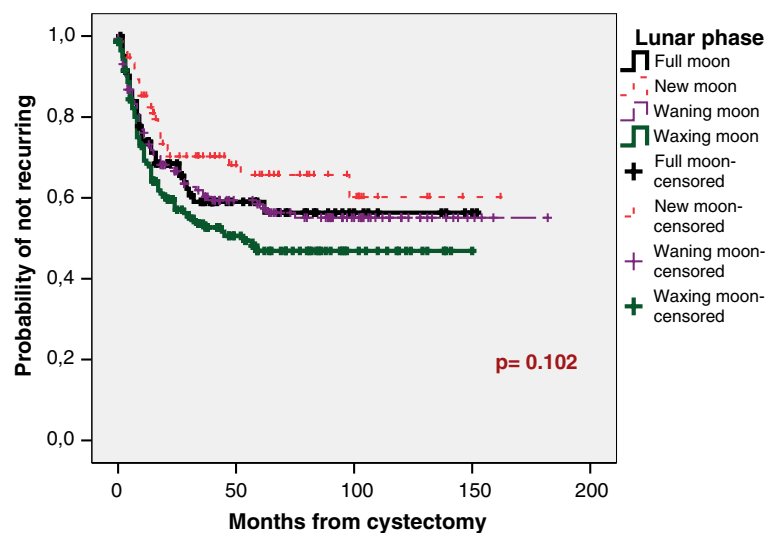
Fig. 1 Progression-free survival in all patients for each of the pathological subgroups (organ-confined, non-organ-confined, pN+)**Fig. 2** Progression-free survival in all patients according to the lunar phase (full moon/ LP1, waning moon/LP2, new moon/LP3, waxing moon/LP4)

Fig. 3 Progression-free survival in all patients according to the zodiac sign (Libra versus non-Libra)

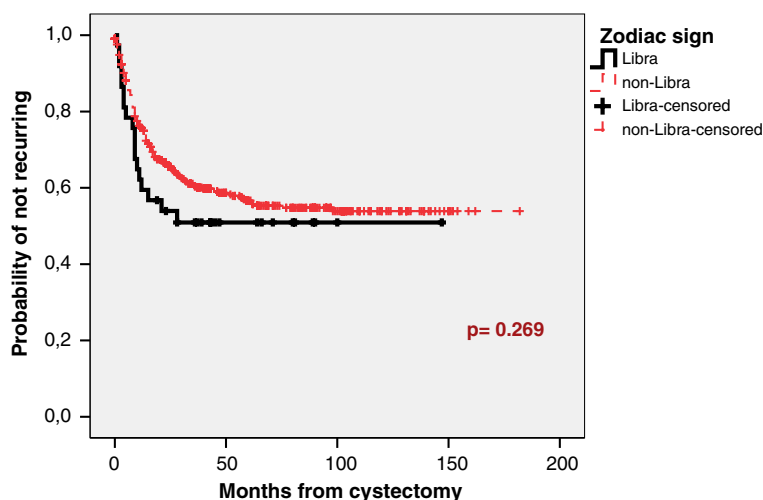
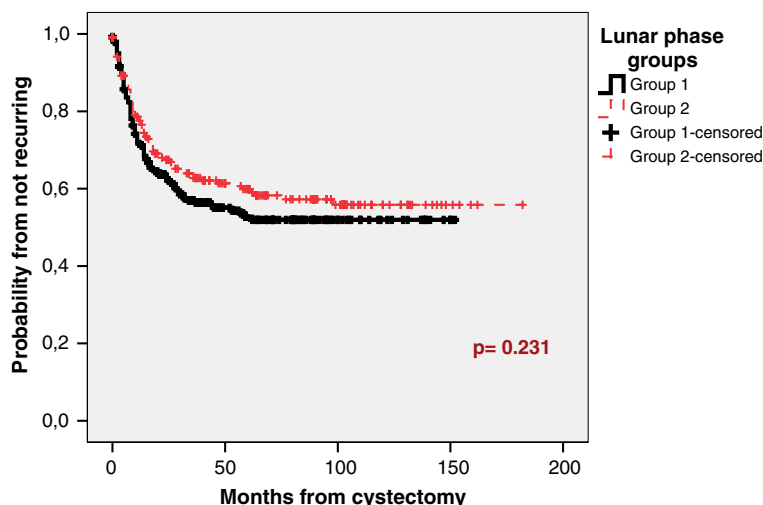


Fig. 4 Progression-free survival in all patients according to the lunar phase groups (group 1/ unfavorable period versus group 2/auspicious period)



hazards model. Multivariate analysis identified the pathological tumor stage to be an independent predictor of progression-free survival (RR = 2.254; 95% CI, 1.90–2.68; $P < 0.001$), whereas the lunar phase group failed to be prognostically significant (RR = 1.139; 95% CI, 0.85–1.52; $P = 0.380$).

Discussion

Circadian rhythm or chronobiology has been investigated in animals and human beings and there is evidence that this phenomenon may influence the immune response [16]. About 10.5% of the German population believe that

the moon cycle influences the outbreak and follow-up of diseases; there is a regional difference with a preference in the southern part of Germany where up to 17.9% trust the moon hypothesis [11]. Apart from the lunar phase as such, the sign of the zodiac applicable to the organic systems to be operated on is deemed to be of importance by proponents of this theory for the post-operative progress [2]. This attitude is based on a dictum from Hippocrates: “Never touch that part of the body with iron that is governed by the sign of the zodiac that is currently crossing the moon” [11]. The topical recommendations find their scientific basis in a report from Andrews, a surgeon from Tallahassee, Florida, USA (1960) who observed an

increased incidence of bleeding complications after tonsilectomy at full moon [17]. However, the data have been re-analyzed—in the original report there is no statistical analysis—and no relationship between moon phase and bleeding complication could be detected [11]. There are only a limited number of publications currently available that evaluate the post-surgical progress according to the lunar phases. At the University Graz, Austria nearly 15,000 consecutive inpatients with surgical procedures including abdominal, vascular, cardiac, thoracic, plastic, and orthopaedic operations showed no relationship between perioperative mortality and lunar phase [8]. Holzheimer et al. also found no correlation between lunar phase and complications after analyzing 782 outpatients (herniotomy, hemorrhoidectomy, or crossectomy) [9]. In one study by Peters-Engl et al. from the Lainz Medical Center, Vienna, Austria lunar phase analysis of 3,757 patients with breast cancer after surgery failed to show a correlation with disease survival [10]. At present there is just one urological article that considers the effect of the lunar phase, even though this has nothing to do with the surgical outcome. Exadaktylos et al. evaluated the influence of lunar phase in 214 patients who were diagnosed with acute urolithiasis [7]. In general, the forces of lunar phase were negligible and not strong enough to be a physiological stimulus and cause renal colic.

The present study shows that timing of surgery during the lunar phase does not influence the subsequent survival and the complication rate of bladder cancer after radical cystectomy. This article is the first report addressing the question of whether the lunar phases have an impact on bladder cancer outcome. Although our study is retrospective, the large number of patients minimizes the risk of usual biases and underlines the magnitude of this analysis. One bias might have affected accurate estimation on complication and survival assessment. Possibly, patients might have selected the date of surgery according to particular lunar phases. However, we can assume that patients were randomly assigned surgery dates and when evaluating our data it was also possible to exclude a lack of representation during asserted unfavorable lunar positions.

The discussion of lunar phase may have other socio-economic aspects. Hospital organization is already under pressure and demands to allocate resources with regard to lunar phase may have detrimental effects. With regard to the anxiety of patients faced with the necessity of an elective operation it has been criticized that patients are unnecessarily put into a state of uncertainty when planning the right time of operation. Additionally, the results from various authors indicated that extending the time window between diagnosis of muscle invasion and radical cystectomy diminished progression-free survival [12, 13, 14]. It would therefore be a welcome step if the book publishers of this type of literature were able to decide to include a reference in the next new editions, that scientific investigations of the presumed effects by lunar phase authors regarding surgical complications have not been confirmed.

In summary, we seem to have evidence that the timing of radical cystectomy during the lunar phase has no impact on the complications and the oncological outcome of bladder cancer. Our results do not support any recommendations for scheduling patients for surgery at any particular day or period of the lunar phase and the zodiac sign.

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